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ECG AND JVP

Electrocardiography (ECG)

00:00:38

ECG LEADS

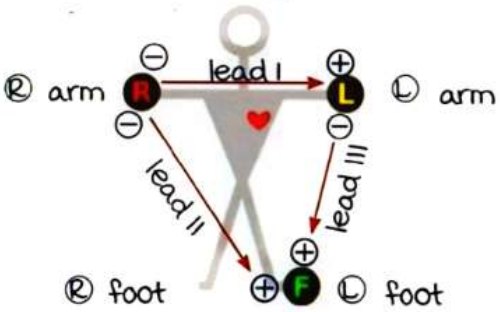
- measures potential difference b/w ⊕ & ⊖ electrodes which causes deflections
  → Represented as waves in ECG.

- 12 leads
  - 6 limb leads
    - Bipolar
      - Lead I.
      - Lead II.
      - Lead III.
    - Unipolar (Augmented)
      - aVR.
      - aVL.
      - aVF.
  - 6 chest leads
    - V1 – V6.

Limb leads :

Bipolar leads :

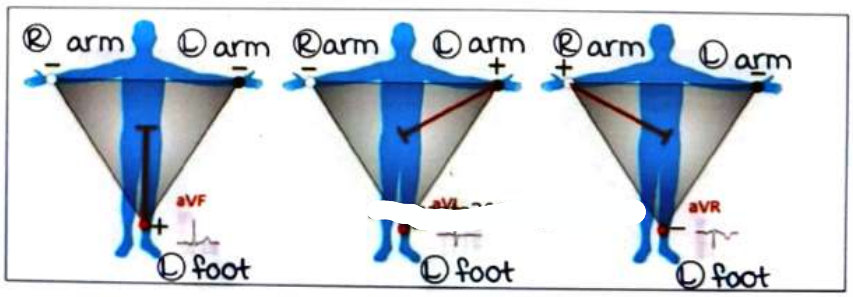
- Einthoven's triangle



- Einthoven's law : Electric potential of lead II = lead I + lead III.

Augmented unipolar leads

- AKA goldberger's leads
- One limb : ⊕
- Remaining 2 limbs : ⊖

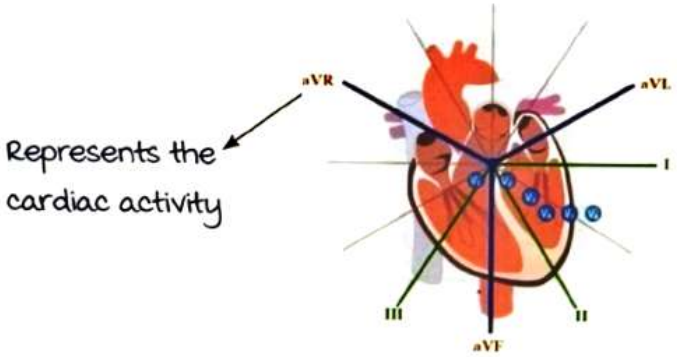


Potentials augmented by 1.5 times

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Clinical significance :

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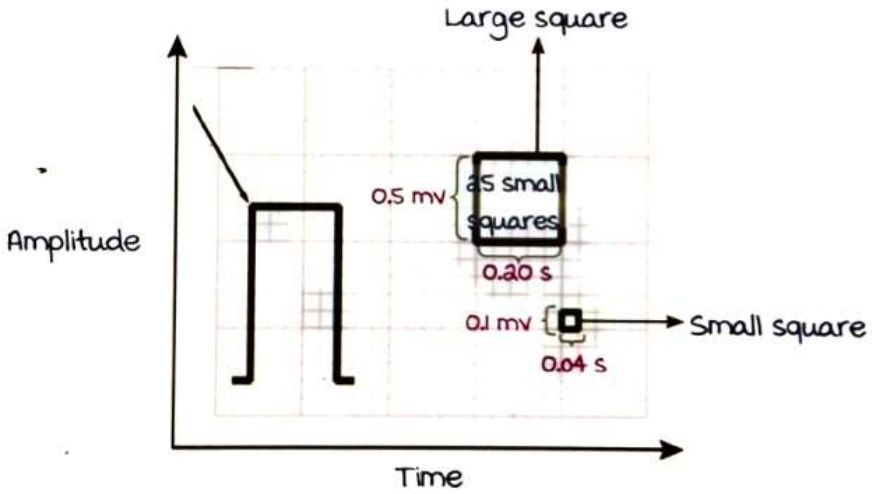


Leads	Surface	Blood supply
V3, V4	Anterior wall	Left Anterior Descending Artery (LAD)
V1, V2	Septal wall	Left Anterior Descending Artery (LAD)
aVL, I, V5, V6	Lateral wall	Left Coronary Artery (LCA) – Circumflex Branch
II, III, aVF	Inferior wall	Right Coronary Artery (RCA)

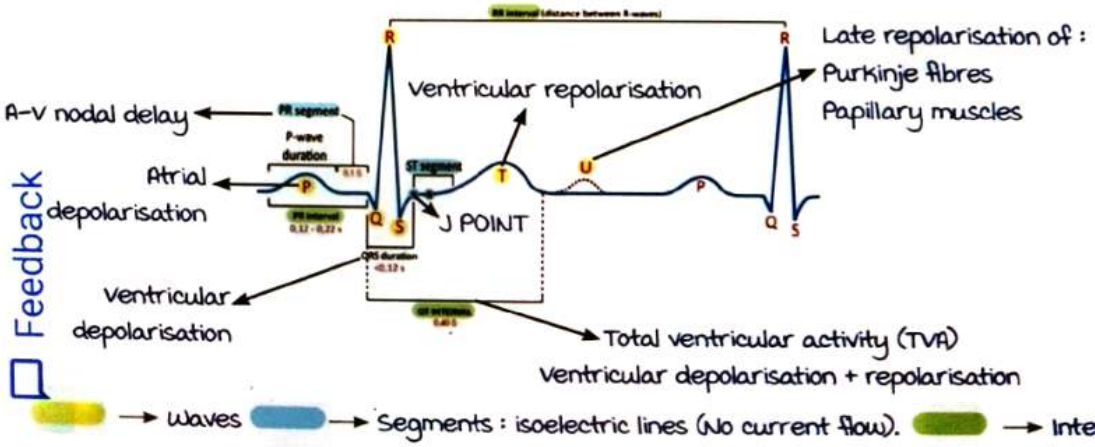
widow maker's artery (m/c artery causing MI)

NORMAL ECG

ECG paper:

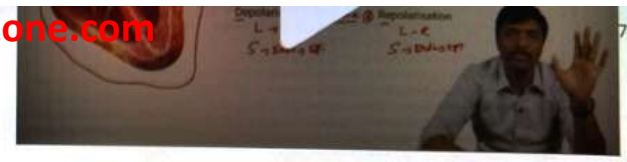


Waveform :



Feedback

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Segment/interval	Applied aspect
ST segment : Elevation/depression	Detects ischemic/infarcted tissue (D/t current flow)
PR interval	>0.20s → heart block
QT interval	Changes with heart rate, therefore Bazett's formula: QTc (Corrected QT) = $\frac{\text{QT interval}}{\sqrt{\text{RR interval}}}$
RR interval	Used to calculate heart rate (HR) HR = $\frac{1500}{\text{no. of small squares b/w 2 successive R waves}}$

Cardiac axis :

Direction of current flow :

- Whole heart :

Ⓛ ⟵ Repolarisation ⟵ / ⟶ Depolarisation ⟶ Ⓡ

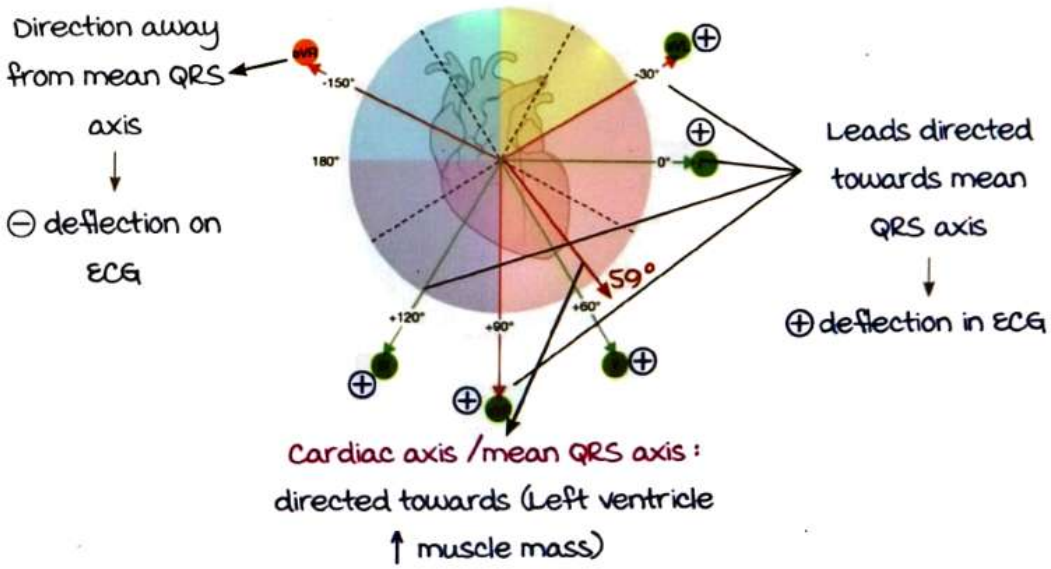
Septum ⟷ Endocardium ⟷ Epicardium

Individual myocardial cell :

Ⓛ ⟵ Repolarisation ⟵ / ⟶ Depolarisation ⟶ Ⓡ

Septum ⟶ Endocardium ⟶ Epicardium

Direction of overall electrical activity :



Cardiac axis /mean QRS axis : directed towards (Left ventricle ↑ muscle mass)

Feedback

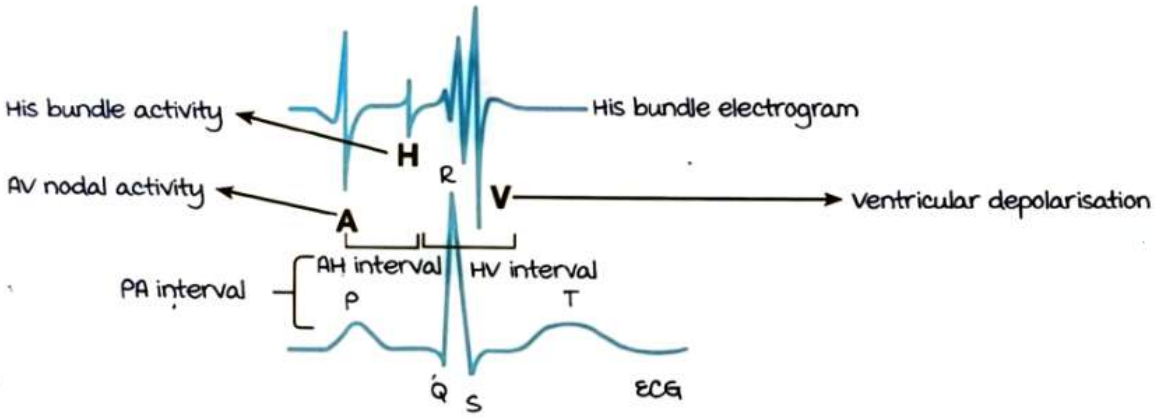
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HIS BUNDLE ECG

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Recorded along with routine ECG.

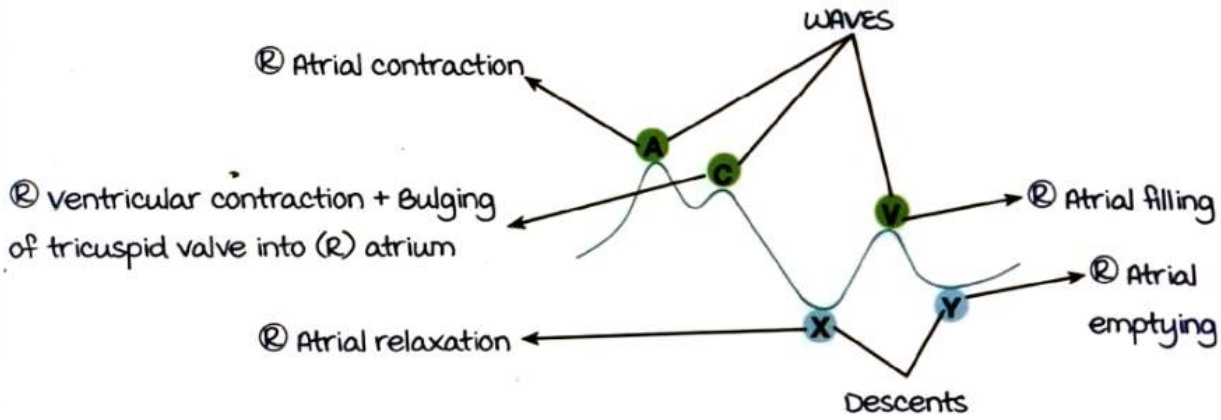
Significance : Cardiac conduction blocks → Intervals prolonged.



Jugular Venous pressure (JVP)

00:43:46

Recorded in internal jugular vein.

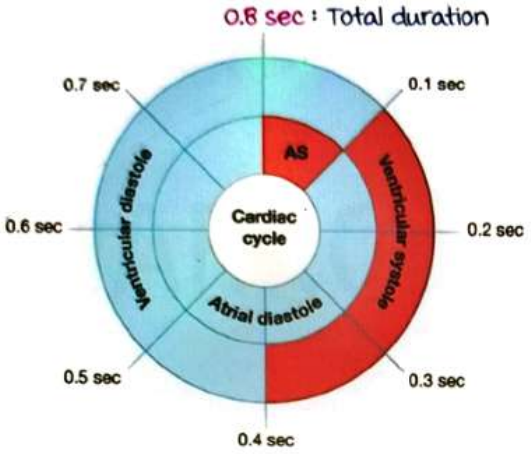


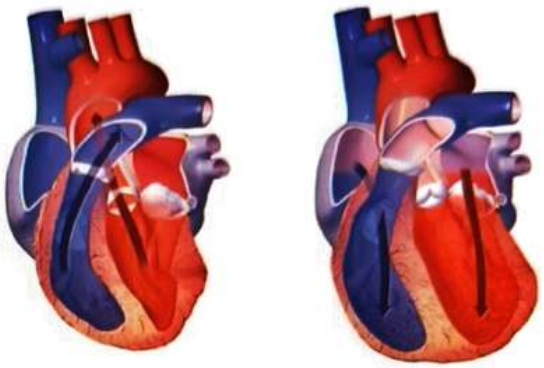
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CARDIAC CYCLE





Systole (pumping/ejection) - Diastole (filling)

Systole
Diastole

Atrial systole : 0.1s
Atrial diastole : 0.7s
Ventricular systole : 0.3s → 3 phases
Ventricular diastole : 0.5s → 5 phases

Clinical aspect :

- Exercise → ↑ HR → ↓ Cardiac cycle (0.3s).
  - → Ventricular systole : 0.12s
  - → Ventricular diastole : 0.18s

  } Ventricular filling compromised.

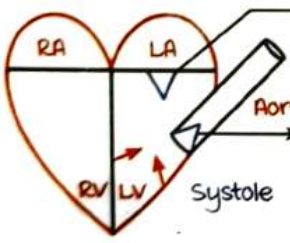
- Cardiac cycle = $\frac{60}{HR}$ $\left(CC \propto \frac{1}{HR}\right)$

Phases of cardiac cycle

00:06:09

VENTRICULAR SYSTOLE :

1. Isovolumetric ventricular contraction :

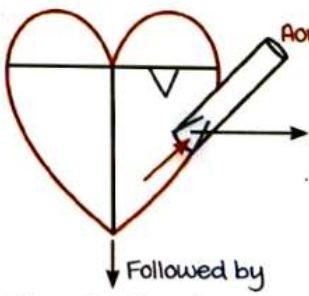


→ mitral valve closes
(1st heart sound (S1) heard)
Aorta
→ Already closed aortic valve
Systole
Volume : same
Pressure ↑↑

At end of phase
Aortic valve opens

2. Rapid ejection phase :

2/3rd of ventricular volume



→ Aortic valve opened

↓ Followed by

3. Slow ejection phase :
Around 1/3rd ejected

↓

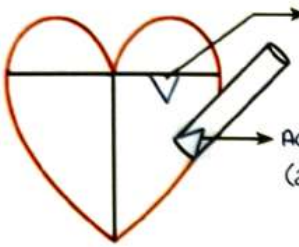
End systolic volume = 50ml.

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VENTRICULAR DIASTOLE :

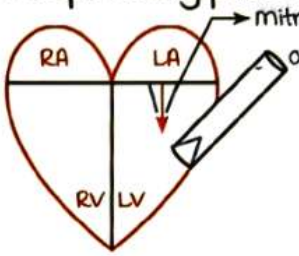
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1. Protodiastole :



2. Isovolumetric relaxation
Volume : Same
Pressure ↓

3. Rapid filling phase :



Passive (80%)

4. Slow filling phase :
(AKA Diastasis)

5. Filling d/t atrial contraction :
20% Active ventricular filling
(Atrial Kick)

End diastolic volume = 120ml

Heart Sounds

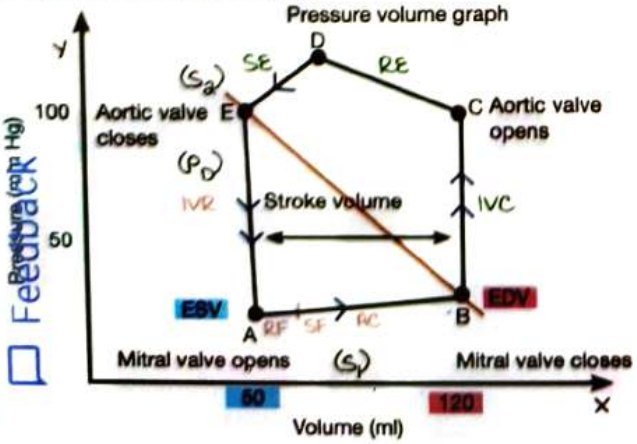
00:18:44

Heart Sound	Etiology	Seen in
S_1	Closure of mitral & tricuspid valve.	Physiological
S_2	Closure of aortic & pulmonary valve.	Physiological
S_3	Vibration of ventricular wall during 3rd phase of diastole (Rapid filling)	Physiological : Children & following exercise
S_4	Stiff ventricles → Forceful atrial contraction (>20% filling)	Pathological : Hypertension & restrictive cardiomyopathy

Pressure Volume Changes (PVC)

00:23:44

NORMAL PHYSIOLOGY



Systolic phase
IVC : Isovolumetric Contraction
RE : Rapid Ejection
SE : Slow Ejection

Diastolic phase
IVR : Isovolumetric Relaxation
RF : Rapid Filling
SF : Slow Filling
AC : Atrial Contraction
PD : Protodiastole

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Formulas :

1. Stroke volume (SV) : End diastolic volume – End systolic volume.
   (EDV) (ESV)
   = 120 – 50 = 70mL

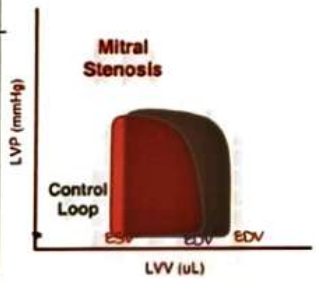
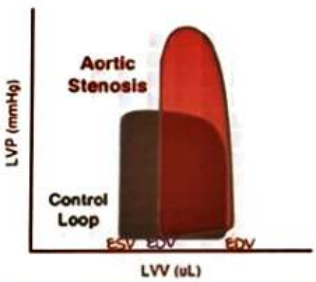
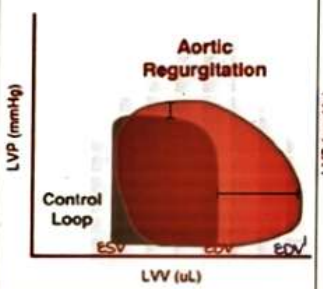
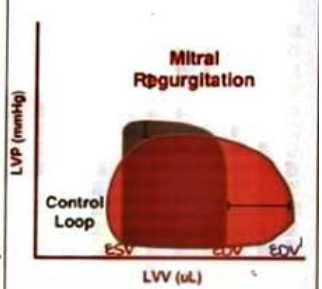
2. Ejection fraction (EF) : $\frac{SV}{EDV} \times 100 = \frac{70}{120} \times 100$

   Normal EF = 55 – 60%

3. End diastolic volume (EDV) : $\frac{ESV}{1-EF}$

4. End systolic volume (ESV) : EDV (1 – EF).

CARDIAC PATHOLOGIES :

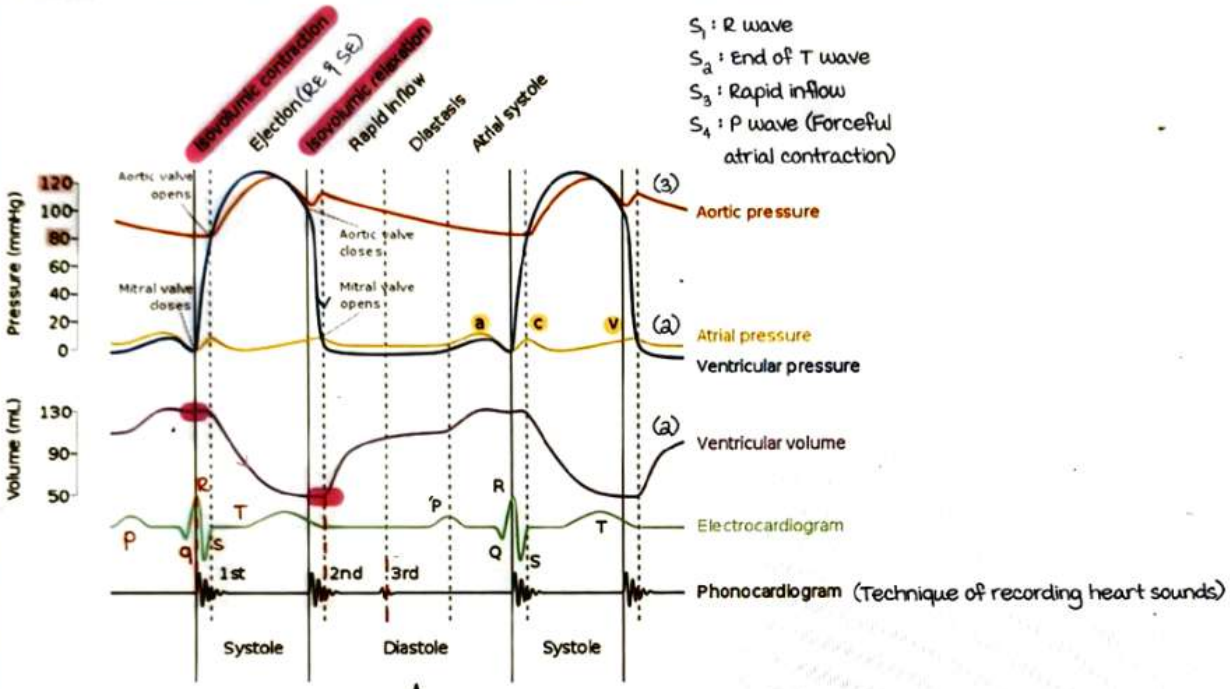
	Mitral Stenosis	Aortic Stenosis	Aortic Regurgitation	Mitral Regurgitation
Pressure Volume alteration	Left shift ← 			
EDV	↓	Normal	↑↑	↑↑
ESV	Normal	↑↑	Normal	–
Other features	↓ stroke volume	↑ left ventricular pressure	• Isovolumetric phase not straight line. • Height↑, width↑.	• Isovolumetric phase not straight line. • Height↓, width↑.

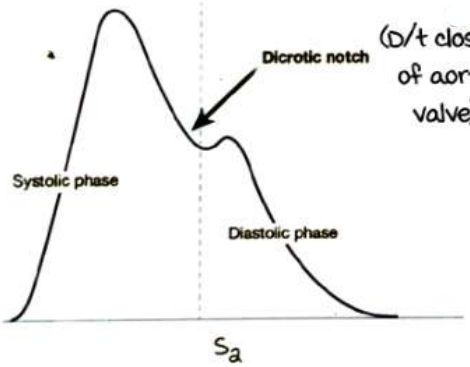
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Wiggers Diagram

00:45:39 ----- Active space -----



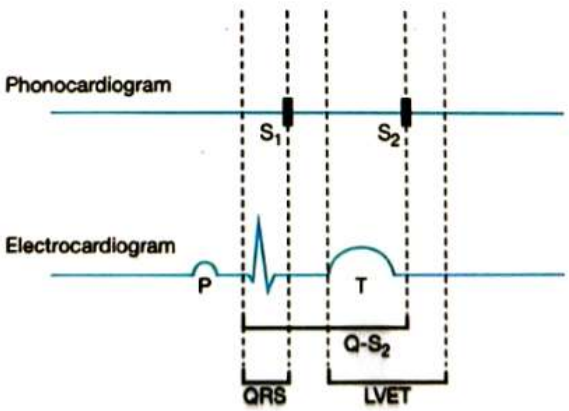


Aortic pressure waveform

Terminologies :

1. Total Electromechanical Systole ($Q-S_2$) :
   - Onset of the QRS complex to the closure of aortic valves (S_2).
2. Left Ventricular Ejection Time (LVET) :
   - Beginning of the carotid pressure rise to dicrotic notch.
   - Correlates with ejection fraction.
3. Preejection period (PEP) :

Difference b/w QS_2 & LVET.



Feedback

$\frac{PEP}{LVET} = 0.35$; ↑ ratio ∝ $\frac{1}{\text{Ejection fraction}}$